



Streamlining Software Delivery with Tekton Pipelines

An open source project, Tekton is a cloud native solution and Kubernetes-native resource for building, testing, and deploying applications. It runs inside Kubernetes or its other variants, enhancing the efficiency of DevOps.

DevOps and Kubernetes (often referred to as DevOps with Kubernetes) complement each other to enable efficient and scalable software development, deployment, and management in modern cloud-native environments.

The combination of Kubernetes container orchestration capabilities and DevOps practices leads to improved collaboration, shorter development cycles, and enhanced application performance in modern cloud-native environments. By leveraging Kubernetes in a DevOps workflow, organisations can achieve faster, more reliable, and scalable software delivery.

Kubernetes offers several advantages when integrated with DevOps practices.

Agility and speed: Faster application deployments and updates enable swift responses to market demands and changes.

Scalability: Automatic horizontal scaling handles varying workloads efficiently.

Reliability and resilience:

Self-healing capabilities and node management ensure continuous application availability.

Consistency and portability:

Abstracting infrastructure allows consistent deployments across diverse environments.

Infrastructure as Code (IaC):

Kubernetes resources defined as code enable version-controlled and repeatable infrastructure management.

CI/CD integration:

It seamlessly integrates with CI/CD pipelines, streamlining the software delivery process.

GitOps:

Version-controlled configurations ensure reliable and consistent application state management.

Microservices and service mesh:

Suitable for microservices architectures, enhanced by service mesh for observability and security.

Resource utilisation:

Optimises resource allocation, leading to cost savings and efficient cloud usage.

Observability:

Extensive monitoring and logging capabilities provide insights into application and infrastructure health.

Multi-cloud and hybrid cloud

support: Cloud-agnostic nature enables seamless deployments across various platforms.

Collaboration and communication:

Encourages cross-functional collaboration, allowing developers to focus on application logic while operations handle infrastructure.

Problems of DevOps in Kubernetes

DevOps in Kubernetes comes with its own set of challenges due to the complexities of managing containerised environments and orchestrating microservices. Here are some common problems faced in DevOps with Kubernetes.

Steep learning curve: Kubernetes has a steep learning curve, especially for teams new to container orchestration and cloud-native technologies.